

Chapter 23

Interactions between lexical tone and intonation in Rìkpa'

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This paper investigates how lexical tone interacts with intonation in Rìkpa', a Cameroonian Bantu language. We present an acoustic analysis of intonation patterns of declarative sentences and polar questions, concluding that questions are produced with overall higher pitch, and that a right edge H% boundary tone is present, which leads to pitch raising at the right phrasal edge in questions. This boundary tone also interacts with lexical tone in key ways: while the boundary tone overrides and augments the pitch of a lexical high tone at the right edge of the sentence, there is little to no effect of raising for right-edge low tones.

1 Introduction

Though the tonal systems of African languages have long been of interest to phonologists, until recently, work on the intonation systems of these languages has been relatively sparse. Recent work has shown that many African tonal languages have robust intonational systems which interact with a range of grammatical properties, such as sentence type (e.g. questions vs. statements, [Chai et al. 2022](#), [Downing 2017](#), [Gjersøe et al. 2016](#), [Kula & Hamann 2017](#), [Patin 2017](#), [Rialland 2007](#), [Rialland & Aborobongui 2017](#)); focus ([Inkelas 1998](#), [Kügler 2017](#), [Kisseberth 2017](#), [Patin 2017](#)) and syntactic boundaries ([Chai et al. 2022](#)). When it comes to marking questions, intonation can be seen to have an effect at either the local or global level (or both). Global intonation involves overall raising or lowering of pitch, a phenomenon which can also occur in conjunction with vowel raising/lowering and phonation changes, a host of properties sometimes

characterized in terms of ‘tense’ vs. ‘lax’ prosody (Lindsey 1985, Rialland 2007, Downing & Rialland 2017). Local intonation involves raising or lowering of pitch on a specific word. When these effects are relegated to the left or the right edge of the phrase, they are typically seen to be caused by a boundary tone. There is variation across African tonal languages in terms of whether and which boundary tones are employed and which syllables they associate to. For example, a right-edge boundary tone might associate with the final syllable, penultimate syllable, or multiple syllables (Rialland 2007 and references therein). Boundary tones may also only surface with specific lexical tones. For example, a boundary tone may surface only when a final lexical high tone is present, but not in the absence of a high tone (Patin 2017). Here, we investigate both global and local properties of intonation in Rìkpá’ to see how the language fits into the general typology of African intonational systems

2 Tone and intonation in Rìkpá’

Rìkpá’ (A53) is a Bantu language spoken by approximately 60,000 speakers in the Center region of Cameroon. The language has SVO word order. Guarisma Popineau (1992) represents one of the most extensive descriptions of the Rìkpá’ language, including the tonal system of Rìkpá’, which includes both high and low, as well as contours which can be formed through the association of two distinct level tones to a single syllable (1).

- (1) a. ká “look for”
- b. kà “serve”
- c. kà + PFV → kǎ “serve (perfective)”
- d. kpâ “speech”

Guarisma (2000) describes some aspects of the intonation of the language, including downstep and downdrift within declarative utterances. As can be seen in Figure 1 (pitch tracks based on the present study), high tones become progressively lower throughout the sentences, whether or not an intervening low tone occurs.

Beyond the effects of these downtrends, there appear to be additional intonational processes in Rìkpá’ that are at play in marking sentence type. Specifically, there are clear intonational differences in the production of declarative statements versus polar questions even where the two constructions are otherwise morphologically identical. Figure 2 and 3 contain two declarative statements,

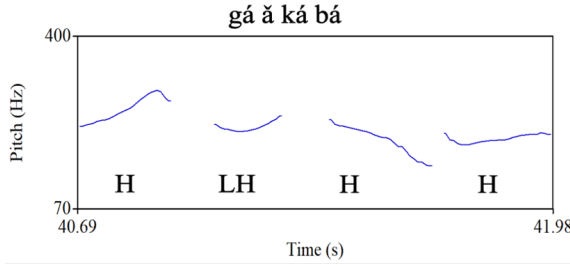


Figure 1: Downdrift in Rìkpá' *gá ă ká bá* 'Mother looked for the meat'

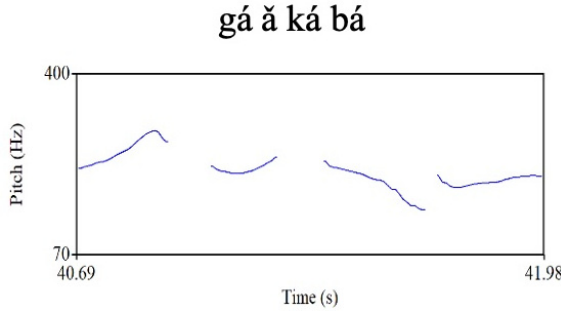


Figure 2: Declarative with H tones

with Figure 2 containing a high subject, verb, and object, as well as a rising tone subject marker in second position (*gá ă ká bá* 'Mother looked for the meat') and Figure 3 containing a low subject, verb, and object (*gá ă ká bá* 'Ga served the sea salt'). Figure 4 and 5 contain corresponding polar questions (e.g. *gá ă ká bá?* 'Did mother look for the meat?') with the same lexical tone melodies. Note that the end of the high tone polar question in Figure 4 has a steep rise in final position, whereas the low tone polar question does not show as much of a rise.

The difference in patterning of the final word across Figure 2-5 suggests that a boundary tone may be present in at least some contexts, e.g. where the sentence ends with a word with a lexical high tone. We now aim to provide an acoustic analysis of f0 in Rìkpá' in order to establish 1) whether or not there is any evidence for the presence of a similar high boundary tone in questions ending with a lexical low tone; 2) whether the boundary tone is limited to occur in constructions which end with a noun, like the examples above; and 3) whether Rìkpá' also shows evidence for more global intonation patterns, such as sentence-level pitch raising in polar questions.

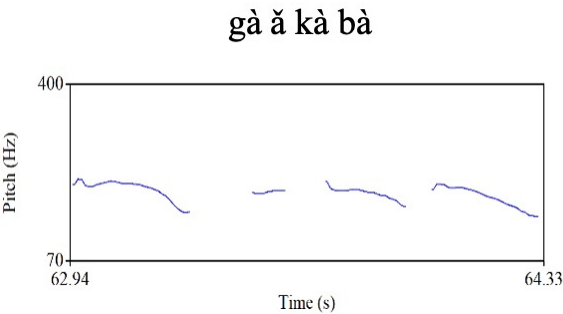


Figure 3: Declarative with L tones

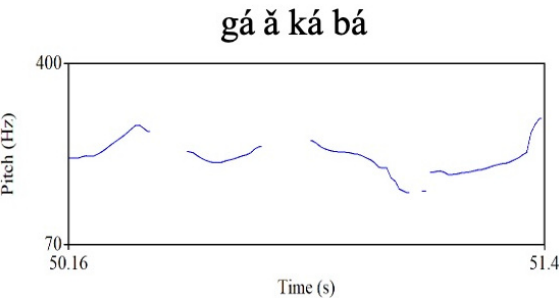


Figure 4: Polar question with H tones

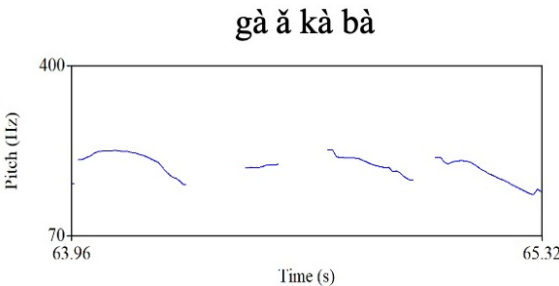


Figure 5: Polar question with L tones

3 Method

Data for this study come from a corpus of pairs of declaratives/yes-no questions in which the subject, verb, and object varied in terms of whether they carried a high or low lexical tone. The structure of the sentences was Subject – Subject Marker – Verb – Object. The subject marker in these sentences always had a LH rising tone. There were two different lexical items used for the subject and object

for each tone, giving a total of 3 (sentence positions) $\times$ 2 (tones) $\times$ 2 (sentence types) $\times$ 2 (subject lexical items) $\times$ 2 (object lexical items) = 48 items. Sentences were repeated 3 times each and recorded in random order for each of six speakers, 3 identifying as male and 3 identifying as female. Sentences were recorded and annotated by the first author in Praat, and then pitch values were extracted from individual phones for analysis. F0 was z-scored by subject in order to control for individual variation in pitch. Linear mixed effects models including fixed factors of Word (subject, verb, or object), Lexical Tone (high or low), and Phrase Type (question or declarative), as well as a by-subject random intercept, were used to evaluate how these variables influenced fundamental frequency. Interactions between these variables were also examined. Main effects for each variable were assessed through a Type III Anova using Satterthwaite's method via the `lmer` test package in R.

4 Results

Table 1 shows the average pitch values (in Hz) for each word in the sentence by lexical tone and phrase type. Overall, the values for words in questions are higher than those in declarative statements.

Table 1: Average f0 (in Hz) for each word by tone and phrase type

Phrase Type	Subject	S.M.	Verb	Object
Declarative L	168.94	160.82	162.36	158.58
Question L	171.05	165.1	166.39	169.22
Declarative H	205.17	188.89	183.92	181.83
Question H	212.47	196.01	221.34	225.90

We see differences in pitch across all words in the sentence by sentence type: polar questions are realized with higher pitch than declarative sentences. Of note, in particular, is the difference in pitch on the last word. We can see there is an especially large difference in pitch between the declarative and polar question for the final word when it carries a high tone. Here, there is an average difference of 44 Hz between conditions. For final low tones, this difference is only 10 Hz. The z-scored results for each word by tone and sentence type are shown in Figure 6.

Results of the linear mixed effects model are shown in Table 2. There was a main effect of Lexical Tone, as expected: high tones were, on average, higher than

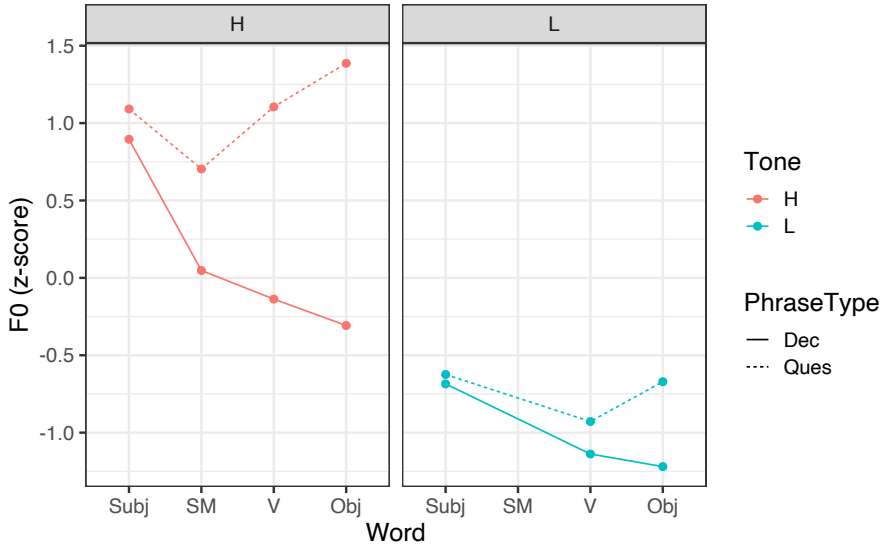


Figure 6: Pitch results by Word, Tone, and Phrase Type

Table 2: Results of statistical analysis

PhraseType	$F(1, 31988) = 889.01$	$p < .001$
Lexical Tone	$F(4, 31988) = 116.74$	$p < .001$
Word	$F(4, 31988) = 61.20$	$p < .001$
PhraseType * Lexical Tone	$F(4, 31988) = 87.17$	$p < .001$
PhraseType * Word	$F(3, 31988) = 17.20$	$p < .001$
PhraseType * Lexical Tone * Word	$F(3, 31988) = 3.78$	$p < .001$

low tones. There was also a significant effect of Phrase Type, with questions generally produced at a higher pitch than statements. There was also a main effect of Word, with words earlier in the sentence realized with higher pitch values than those later in the sentence. A significant two-way interaction between Phrase Type and Lexical Tone reflected the fact that the effect of question raising was greater for words with lexical high tones than with low tones. An interaction between Phrase Type and Word reflected that declination effects seen in declarative sentences were not as pronounced – or even reversed – in questions, where the effects of the boundary tone can be seen to raise the pitch of the phrase-final word to a higher level than the initial word of the sentence. Finally, the three-way interaction between Phrase Type, Lexical Tone, and Word reflected the fact

that differences in pitch across phrase type and sentence position were more pronounced where the underlying tone of the words was high than when they were low.

5 Follow-up analysis: Sentences ending in adverbs

So far, our analysis has only taken into account sentences of a specific structure, i.e. those involving a subject, subject marker, transitive verb, and an object. In order to explore whether the effects of the boundary tone can be seen more generally, we had participants record new stimuli which included the same material as the previous sentences, plus one of two additional adverbs, *gênÁ*, ‘today’ which has a HLH melody, and *dîŋko*, ‘yesterday’ which has a LL melody. We followed the same procedure for the prior dataset.

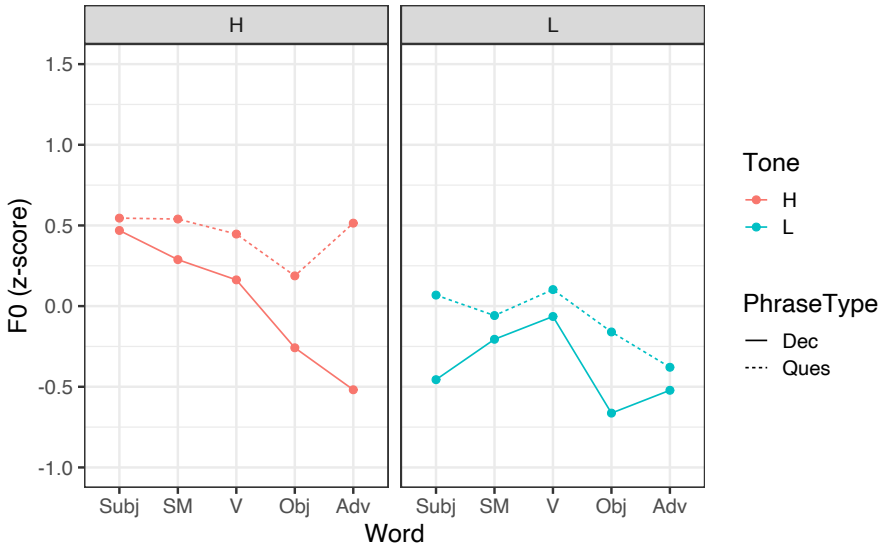


Figure 7: Pitch results by Word, Tone, and Phrase Type, with adverbs included

Once again, we see an overall higher pitch level for questions as compared with declaratives. As before, we see a rise on the final word of the sentence for high tone words within questions, consistent with the presence of a boundary tone. We do not see the same rise on the last word for the low tone words in questions, though the difference in pitch between Phrase Type conditions does appear to increase somewhat at the object position.

We can now take a closer look at the two adverbs uttered in the two phrase type conditions to see exactly where the boundary tone is taking effect. In Figure 8, the x-axis represents the timepoint within the adverb as a percentage. We can see that the that the pitch rises toward the last third of the word and then plateaus, indicating the boundary tone likely docks to the last high tone syllable. For the adverb with the LL melody, we again do not see the same strong evidence for a boundary tone, though there is a very slight rise between the last two timepoints for questions.

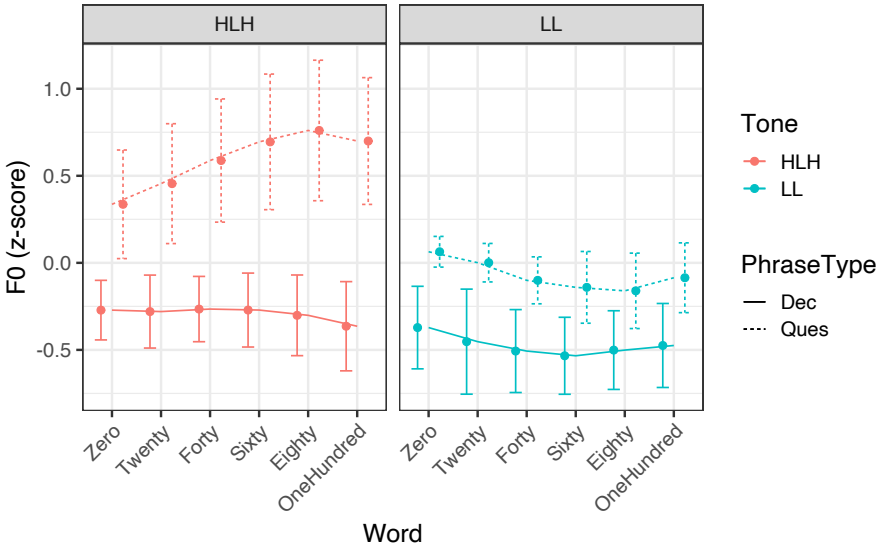


Figure 8: Pitch patterns on final adverb by Phrase Type and Timepoint

Finally, we consider data for sentences ending with the LL adverb in order to see how the tone of the preceding object influences the presence or location of the high boundary tone. In other words, where we have a low tone in final position, do we find evidence that the high boundary tone in questions docks somewhere earlier in the sentence? The answer to this appears to be ‘yes’: whereas the highest pitch in questions is realized on the object (in penultimate position) when it is high, the highest pitch is realized on the verb (in antepenultimate position) when the object is low and the verb itself his high.

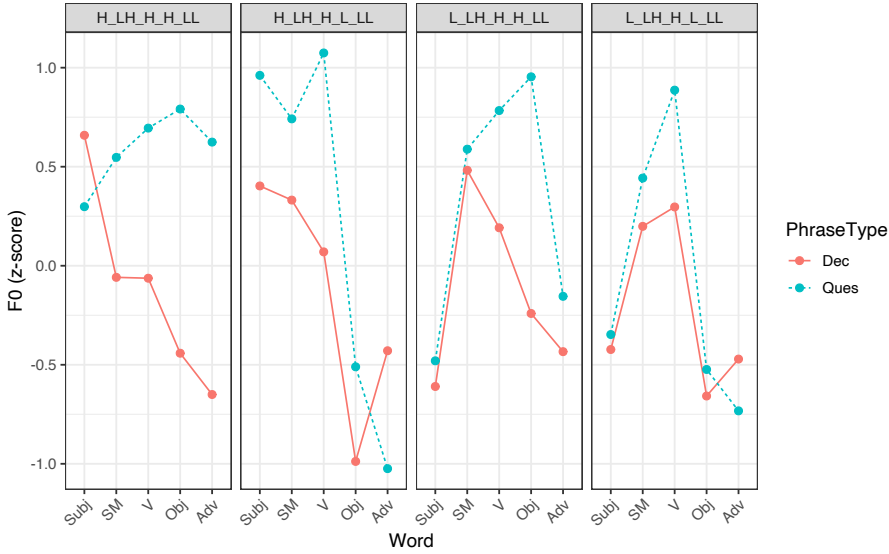


Figure 9: Change in boundary tone location based on location of final H lexical tone (object vs. verb)

6 Discussion

To sum up, Rikpa' shows phonetic effects which are characteristic of tense prosody in polar questions, such as overall higher pitch (Rialland 2007). In addition, we find local effects of pitch raising: a high, or even super-high, boundary tone is present at the right edge, attaching to the final high tone of the utterance and pushing the pitch of that syllable into a higher range than what is expected for lexical high tones. Final low tones show some raising, but not nearly as much as final high tones, and not as consistently. Therefore, although the boundary tone may be present in contexts with phrase-final lexical low tones, it appears to be mostly overridden by the lexical tone of the final word. The fact that there appear to be subtle phrase-final raising effects in at least some contexts (e.g. see Figure 6 for questions ending in an object) suggests that this effect of tonal override may be gradient, rather than categorical, in nature. Another possibility is that speakers might show variability in whether or not they show effects of a boundary tone in low environments. In this case, averaging over all speakers might obscure some more robust patterns of raising for certain speakers. We leave this question for future investigation.

The pattern of boundary tone association at the right edge in Rikpa' is similar to what is described for a range of other Bantu languages from different ar-

eas, including Ekoti (P30), Embosi (C25), Herero (R30) and Kinyarwanda (JD60) (Schadeberg & Mucanheia 2000, Möhlig et al. 2002, Kimenyi 1980). We note that ours is the first study to document this type of pattern in a Bantu language from Zone A. Similar to Shingazidja (G44a), as described by Patin, we find a mix of boundary tone overriding lexical tone vs. being blocked by lexical tone. However, the position and nature of boundary tones (e.g. whether they override or are overridden by lexical tones) is quite variable, and this variation is worthy of deeper investigation. It will also be necessary to investigate the behavior of the boundary tone in Rikpa' in order to understand its patterning in a broader array of sentence types and information structural contexts. A larger survey of the behavior of boundary tones in Bantu languages is also a promising next step in understanding how these tones are represented by speakers and how they interact with lexical tonology.

Finally, the present study has only considered a single quantitative measure – average pitch – in the patterning of different sentence types. This has been a good first step in investigating interactions of tone and intonation in Rikpa', as we see robust differences in average pitch across the conditions of interest. However, our pitch tracks also reveal considerable variation in the overall shape of the pitch contours for different words across tonal and sentential contexts. Future work should focus on how other pitch-related factors such as tonal slope and pitch range may be implicated in the production and perception of lexical tones and boundary tones in Rikpa'.

Abbreviations

H	High lexical tone	PFV	Perfective
L	Low lexical tone	SM	Subject Marker
H%	High boundary tone		

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